

NeuroScan AI v3.0 — Clinical Imaging Report

AI-assisted neuro-radiological screening (research use only)

Report ID	NS3-20260913-093731	Generated	13 Sep 2026 09:37
Source	brisc2025_test_00001_gl_ax_t1.jpg	Input domain	full_image
Deployed arm	full_image	Backbone	EfficientNet-B0

PRIMARY AI IMPRESSION

AI impression	GLIOMA
Ensemble confidence	90.09%
Risk stratum	HIGH
Predictive uncertainty	sigma = 0.0233 (Low)
Branch consensus	100% of the three branches agree

CLASS POSTERIOR

Class	Ensemble	Standard	MC dropout	TTA	MC sigma	
glioma	90.09%	90.29%	90.53%	88.76%	0.0381	SELECTED
meningioma	3.03%	2.78%	2.44%	4.82%	0.0172	
no_tumor	2.42%	2.28%	2.33%	2.70%	0.0136	
pituitary	4.46%	4.65%	4.70%	3.72%	0.0244	

Fusion weights 0.00 / 0.75 / 0.25 (standard / MC dropout / TTA) were selected by grid search on the validation split; the held-out test split took no part in choosing them.

LESION GEOMETRY (Attention U-Net)

Area	259 px ² (0.58% of the crop)	Circularity	0.705
Centroid	(131, 172)	Quadrant	Lower-Right

INTERPRETATION

Imaging features are consistent with glioma, a primary intra-axial tumour of glial origin. Irregular enhancement, mass effect and perilesional oedema are the features that typically drive this classification. Urgent neuro-oncology referral and tissue sampling are the standard next steps.

Recommendation: Urgent neuro-oncology referral; MR spectroscopy, perfusion imaging and tissue sampling.

MODEL PROVENANCE (measured, not asserted)

Dataset	BRISC2025 (briscdataset/brisc2025) — 4211 train / 744 val / 995 held-out test
Test split	Official BRISC2025 test folder, untouched during training
Segmentation	Attention U-Net on annotated masks — test Dice 0.8400, IoU 0.7756
Deployed classifier	full_image — test accuracy 98.59%, macro F1 98.62%
With ensemble	test accuracy 98.79%, macro F1 98.84%, kappa 0.9835
Controlled contrast	full-image arm macro F1 98.62% vs ROI arm 97.56%; McNemar p = 0.0433
Explainability	Grad-CAM on the last backbone convolution, computed on the same input domain the model was trained on

Uncertainty	MC dropout, 30 stochastic passes (BatchNorm frozen)
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IMPORTANT: this is an experimental research system. This report is not a medical diagnosis, has not been reviewed by a licensed clinician, and must not be used to direct patient care. Model outputs are derived from a single 2D slice and must be correlated with the full study, clinical history and expert radiological interpretation. The authors accept no liability for decisions based on this output.